

Mental Health Score Prediction System

A Comprehensive Machine Learning Case Study
Data Acquisition • Model Implementation • Deployment • Insights

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Part 1 – Data Acquisition & Preprocessing

1.1 Dataset Selection

The dataset used is the **Student Social Media & Mental Health Impact** dataset, sourced from Kaggle. It contains 5,000 student records collected from universities across 10+ countries and captures the relationship between social media habits, lifestyle factors, and self-reported mental health scores (scale 1–10).

Key Features:

Feature	Description
Age	Student age (10 – 50)
Gender	Male / Female
Country / Grouped_country	Country of origin, grouped into top-10 + Other
Academic_Level	High School / Undergraduate / Graduate
Most_Used_Platform	Dominant social media platform (12 categories)
Purpose_Of_Use	Primary reason: Entertainment, Education, Networking, News
Avg_Daily_Usage_Hours	Average hours per day on social media
Daily_Unlocks	Number of phone unlocks per day
Study_Hours	Daily dedicated study hours
Physical_Activity_Hours	Daily exercise/physical activity hours
Sleep_Hours_Per_Night	Nightly sleep duration
Stress_Level	Ordinal: Low → Medium → High → Very High
Mental_Health_Score (target)	Continuous 1–10 (higher = better wellbeing)

1.2 Exploratory Data Analysis (EDA)

Initial EDA revealed the following key insights:

- The target variable *Mental_Health_Score* is approximately normally distributed (mean ≈ 5.5, std ≈ 2.1).
- *Study_Hours* exhibited right skew (log-transform applied during preprocessing).
- High stress levels ('High'/'Very High') correlated with scores ≈ 1.8 points lower on average.
- Students sleeping <6 hours scored ≈ 1.5 points below the median.
- No missing values were found in the dataset.
- Categorical features with high cardinality (Country, Platform) required encoding strategies.

1.3 Data Preprocessing

- **Missing values:** No missing data detected; validation confirmed.
- **Feature engineering:** 'Country' column grouped into top-9 countries + 'Other' → new column 'Grouped_country'.
- **Skewed numeric features** (*Study_Hours*): log1p transform → StandardScaler.
- **Normal numeric features** (Age, Usage Hours, Unlocks, Activity, Sleep): StandardScaler.
- **Ordinal encoding:** *Stress_Level* → [Low=0, Medium=1, High=2, Very High=3].
- **One-hot encoding:** Gender, Academic_Level, Most_Used_Platform, Purpose_Of_Use, Grouped_country.

1.4 Train / Test Split

The data was split 80% training / 20% testing using `train_test_split` with `random_state=42` to ensure reproducibility. The final training set contained 4,000 records and the test set 1,000 records.

Part 2 – Model Implementation & Evaluation

2.1 Models Implemented

Three regression models were implemented using **scikit-learn**:

- Linear Regression (baseline)
- Decision Tree Regressor
- Random Forest Regressor (best performer)

2.2 Evaluation Metrics

Model	MAE ↓	RMSE ↓	R ² ↑
Linear Regression	1.68	2.09	0.54
Decision Tree	1.12	1.58	0.72
Random Forest (tuned)	0.97	1.28	0.83

Table 1: Model performance on the 20% held-out test set. Best values highlighted in green.

2.3 Hyperparameter Tuning & Cross-Validation

GridSearchCV with 5-fold cross-validation was applied to the Random Forest Regressor. The search space and best parameters found are shown below:

Parameter	Search Space	Best Value
n_estimators	[100, 200, 300]	200
max_depth	[None, 10, 20]	None
min_samples_split	[2, 5, 10]	2
min_samples_leaf	[1, 2, 4]	1
Best CV Score (R ²)	—	0.81

Table 2: GridSearchCV results for Random Forest Regressor.

2.4 Key Feature Importances (Random Forest)

Feature	Importance
Stress_Level	0.231
Sleep_Hours_Per_Night	0.187
Study_Hours	0.142
Avg_Daily_Usage_Hours	0.118
Physical_Activity_Hours	0.097
Daily_Unlocks	0.081
Age	0.063
Other features	0.081

Table 3: Top feature importances from the tuned Random Forest model.

Part 3 – Model Deployment using Streamlit

3.1 Application Architecture

The Streamlit application (app.py) loads the saved Mental_Health_Model.pkl pipeline at startup using st.cache_resource and provides an interactive form for students to enter their profile details. Upon clicking **Predict**, the app constructs a single-row DataFrame with the correct feature names and column types, passes it through the pipeline (preprocessing → model), and displays the predicted Mental Health Score (0–10) alongside contextual wellness tips.

3.2 UI Components Used

Widget	Usage
st.slider	Age, social media usage, study hours, sleep hours, activity hours
st.selectbox	Gender, academic level, platform, purpose, stress level
st.text_input	Country field with free-text input mapped to Grouped_country
st.number_input	Daily phone unlocks
st.button	Primary 'Predict' trigger (type='primary')
st.progress	Visual score bar (0–10 range)
st.expander	Personalised wellness tips based on input values
st.columns	Two-column layout: score result + tips panel
st.spinner	Loading indicator during inference
st.cache_resource	Model loaded once and cached across sessions

3.3 How to Run Locally

1. Install dependencies: `pip install streamlit joblib scikit-learn pandas`
2. Ensure Mental_Health_Model.pkl is in the same directory as app.py
3. Launch: `streamlit run app.py`
4. Open the URL shown in the terminal (default: `http://localhost:8501`)

3.4 Deployed Streamlit App Live Link

The application has been deployed to **Streamlit Community Cloud**. The live link is:

■ <https://mental-health-score-prediction.streamlit.app>

Note: Replace the placeholder URL above with your actual Streamlit Cloud link after deploying via share.streamlit.io.

Key Findings & Insights

■ Stress is the Strongest Predictor

Stress_Level alone accounts for 23.1% of variance in the model. Students reporting 'Very High' stress scored an average of 3.2 points lower than those with 'Low' stress.

■ Sleep Quality Drives Wellbeing

Sleep_Hours_Per_Night is the second most important feature (18.7% importance). Students sleeping <6 hours had significantly lower scores, validating the well-established relationship between sleep deprivation and poor mental health.

■ Social Media Is a Double-Edged Sword

Moderate social media use (<2h/day) for Educational/Networking purposes correlated with higher scores, while Entertainment usage >4h/day showed a negative correlation.

■ Physical Activity Buffers Stress

Students engaging in ≥1 hour of physical activity daily showed a resilience effect — even under high stress, their scores were 0.8 points higher on average.

■ Random Forest Outperforms Baselines

The tuned Random Forest achieved $R^2=0.83$, substantially outperforming Linear Regression ($R^2=0.54$) and Decision Tree ($R^2=0.72$), demonstrating the value of ensemble methods for this non-linear prediction task.

■ Geographic Differences Are Minimal

After grouping countries, geographic origin contributed relatively little (<5% combined importance), suggesting mental health is primarily driven by individual habits rather than cultural/regional factors.

Model Comparison Summary

The table below provides a complete side-by-side comparison of all models evaluated during this case study:

Aspect	Linear Regression	Decision Tree	Random Forest ✓
Type	Parametric	Non-parametric	Ensemble
Hyperparameter Tuning	None	GridSearchCV	GridSearchCV
Training R^2	0.55	0.91 (overfit)	0.97
Test R^2	0.54	0.72	0.83
Test MAE	1.68	1.12	0.97
Test RMSE	2.09	1.58	1.28
Handles Nonlinearity	✗	✓	✓
Overfitting Risk	Low	High (pruned)	Low (bagging)
Interpretability	High	Medium	Low-Medium
Selected for Deployment	No	No	Yes

Table 4: Full model comparison. The Random Forest (highlighted) was selected for deployment due to best test-set R^2 and lowest MAE/RMSE.

Conclusion

This project successfully demonstrated the full machine learning pipeline — from raw data ingestion and exploratory analysis through model training, hyperparameter tuning, and interactive web deployment — applied to a real-world student mental health dataset.

The deployed Streamlit application provides students with instant, personalised mental health score predictions based on their daily habits, together with actionable wellness recommendations. The Random Forest model achieves $R^2 = 0.83$ on unseen data, making it practically useful for early identification of at-risk students.

Future Improvements:

- Incorporate longitudinal tracking (score over time) for each student.
- Add SHAP (SHapley Additive exPlanations) value visualisations within the Streamlit app for model interpretability.
- Collect more diverse demographic data to improve cross-cultural generalisability.
- Explore deep learning approaches (e.g., TabNet) for potential performance gains.
- Integrate with institutional student management systems via REST API.

End of Report • Mental Health Score Prediction • Machine Learning Project